

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> To answer this question, we need to look at the symptoms and the diagnosis of the patient. The patient is diagnosed with community-acquired pneumonia, which is a type of infection that is typically caused by bacteria. The likely cause of the patient's condition is a bacterial infection, such as Streptococcus or Pneumococcus, which is known to trigger a Th1-cell response as part of the immune response. The other options are not as likely to be the correct answer, as TLR5 is typically activated by bacterial flagellin, which is not a typical feature of the pathogens that cause community-acquired pneumonia, and peptidoglycan is a major antigen that is more typically associated with infections caused by Gram-positive bacteria. Encapsulation is a feature of some types of fungi, not the likely cause of this patient's condition.

</think><answer> It primarily induces the Th1-cell response.